

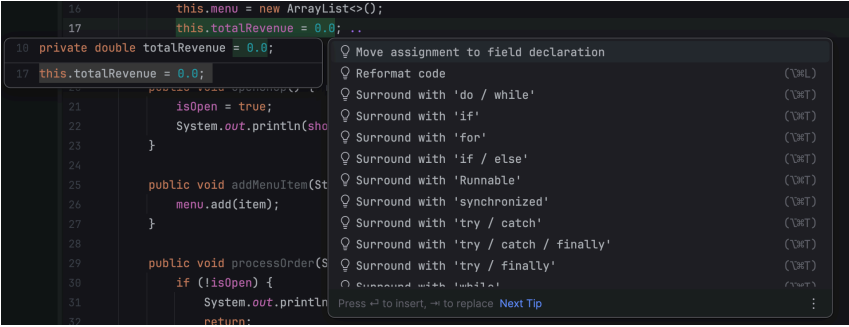
# Test report

| Field    | Content                 |
|----------|-------------------------|
| Feature  | Command Completion (..) |
| Idea     | IntelliJ IDEA 2026.1.1  |
| Language | Java                    |
| OS       | macOS                   |
| Date     | 07.05.2026              |
| Time     | 21:00                   |
| Tester   | Daria Luhova            |

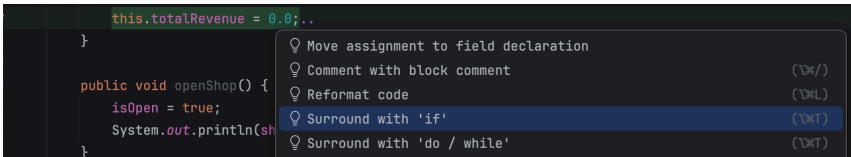
## Testing scope

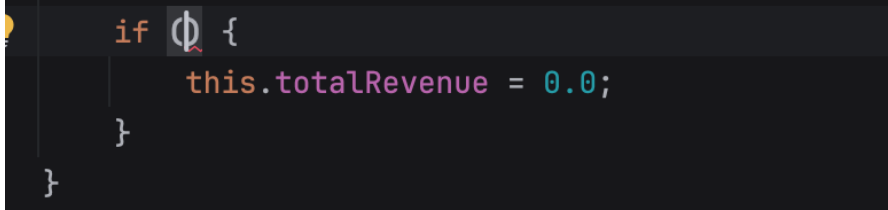
| Area                                                  | Covered |
|-------------------------------------------------------|---------|
| Basic invocation of Command Completion via ..         | YES     |
| Surround actions (if, try/catch)                      | YES     |
| Code transformation (Move assignment, Reformat)       | YES     |
| Block commenting via Command Completion               | YES     |
| Postfix templates (.if, .for) via Command Completion  | YES     |
| AI-powered actions (Explain code, Add Javadoc)        | YES     |
| Complex refactoring (Transform to single exit-point)  | YES     |
| Invocation on method declaration boundary             | YES     |
| Kotlin support                                        | NO      |
| Undo/Redo after commands                              | NO      |
| Invocation inside comments and String literals        | NO      |
| All postfix templates (.null, .nonnull, .return etc.) | PARTIAL |

## Test case 1:

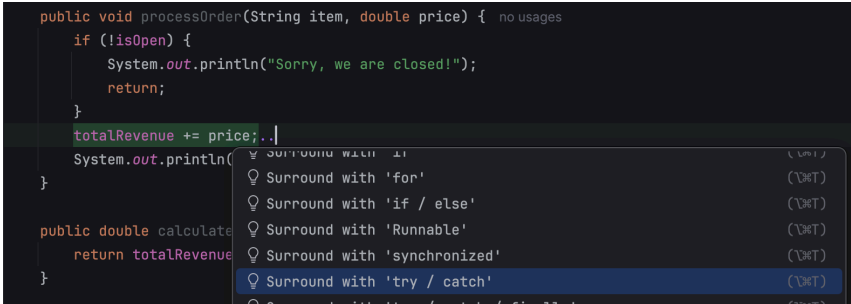
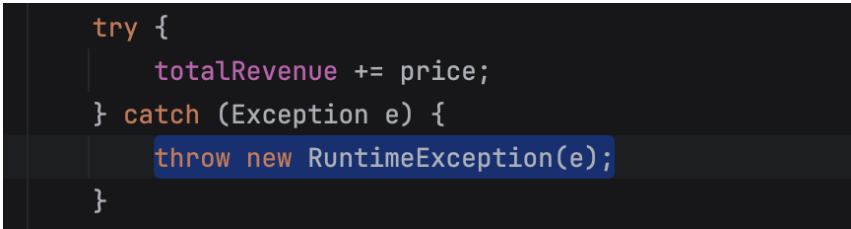
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|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>            | Basic invocation                                                                                                                                                                                                                                                                                                                                      |
| <b>Description</b>     | Verify that Command Completion dropdown appears when .. is typed at the end of a statement                                                                                                                                                                                                                                                            |
| <b>Test data</b>       | this.totalRevenue = 0.0; inside the constructor                                                                                                                                                                                                                                                                                                       |
| <b>Steps</b>           | <ol style="list-style-type: none"> <li>1. Open CoffeeShop.java</li> <li>2. Place cursor at the end of statement <i>this.totalRevenue = 0.0</i></li> <li>3. Type: ..</li> </ol>                                                                                                                                                                        |
| <b>Expected result</b> | Command Completion dropdown appears with a list of context-relevant actions                                                                                                                                                                                                                                                                           |
| <b>Actual result</b>   | <p>Command Completion dropdown appeared with a list of context-relevant actions including Move assignment to field declaration, Reformat code, Surround with 'if', 'do/while', 'for', 'if/else', 'Runnable', 'synchronized', 'try/catch', 'try/catch/finally'</p>  |
| <b>Pass / Fail</b>     | Pass                                                                                                                                                                                                                                                                                                                                                  |

## Test case 2:

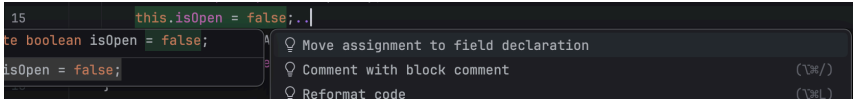
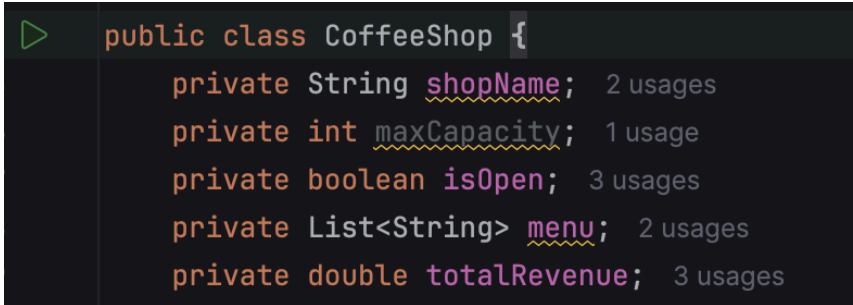
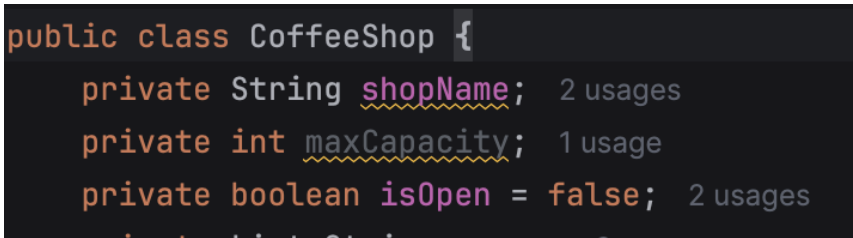
|                        |                                                                                                                                                                                                                                                                                                                  |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>            | Surround with 'if'                                                                                                                                                                                                                                                                                               |
| <b>Description</b>     | Verify that a statement can be wrapped in an if-block via Command Completion and all necessary code is in if statement block                                                                                                                                                                                     |
| <b>Test data</b>       | this.totalRevenue = 0.0; inside the constructor                                                                                                                                                                                                                                                                  |
| <b>Steps</b>           | <ol style="list-style-type: none"> <li>1. Open CoffeeShop.java</li> <li>2. Place cursor at the end of statement <i>this.totalRevenue = 0.0</i></li> <li>3. Type: ..</li> <li>4. Select <i>Surround with 'if'</i></li> </ol>  |
| <b>Expected result</b> | Statement is wrapped in if () { } block, cursor placed inside condition                                                                                                                                                                                                                                          |

|                      |                                                                                                                                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actual result</b> | <p>Statement was placed inside the if block correctly: <code>if () { this.totalRevenue = 0.0; }</code> — cursor placed inside condition ()</p>  |
| <b>Pass / Fail</b>   | Pass                                                                                                                                                                                                                              |

### Test case 3:

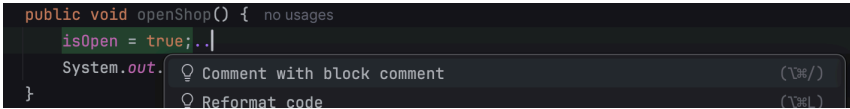
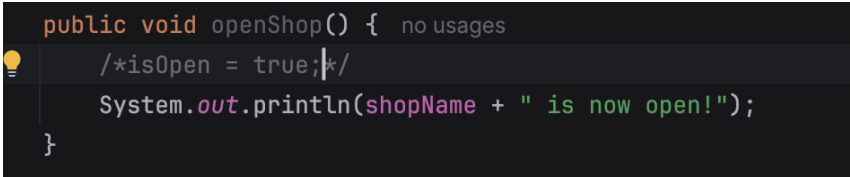
|                        |                                                                                                                                                                                                                                                                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>            | Surround with 'try/catch'                                                                                                                                                                                                                                                                                                                 |
| <b>Description</b>     | Verify that a statement can be wrapped in a try/catch block via Command Completion                                                                                                                                                                                                                                                        |
| <b>Test data</b>       | <code>totalRevenue += price</code> inside <code>processOrder()</code>                                                                                                                                                                                                                                                                     |
| <b>Steps</b>           | <ol style="list-style-type: none"> <li>1. Open <code>CoffeeShop.java</code></li> <li>2. Place cursor at the end of statement <code>totalRevenue += price;</code></li> <li>3. Type: ..</li> <li>4. Select <i>Surround with 'try/catch'</i></li> </ol>  |
| <b>Expected result</b> | Statement wrapped in try/catch block with a relevant exception type                                                                                                                                                                                                                                                                       |
| <b>Actual result</b>   | <p>Statement wrapped correctly: <code>try { totalRevenue += price; } catch (Exception e) { throw new RuntimeException(e); }</code></p>                                                                                                                |
| <b>Pass / Fail</b>     | Pass                                                                                                                                                                                                                                                                                                                                      |

#### Test case 4:

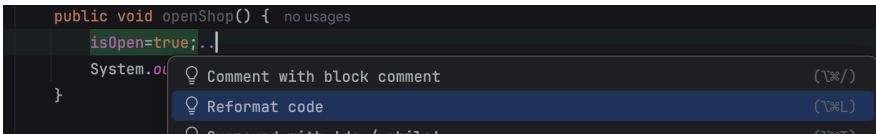
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|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>            | Move assignment to field declaration                                                                                                                                                                                                                                                                                                  |
| <b>Description</b>     | Verify that an assignment in constructor can be moved to the field declaration                                                                                                                                                                                                                                                        |
| <b>Test data</b>       | <i>this.isOpen = false</i> inside constructor                                                                                                                                                                                                                                                                                         |
| <b>Steps</b>           | <ol style="list-style-type: none"> <li>1. Place cursor at the end of statement <i>this.isOpen = false</i>;</li> <li>2. Type: ..</li> <li>3. Select 'Move assignment to field declaration'</li> </ol>                                                |
| <b>Expected result</b> | Line removed from constructor; field updated to private boolean isOpen = false                                                                                                                                                                                                                                                        |
| <b>Actual result</b>   | <p>Line this.isOpen = false removed from constructor; field declaration updated to private boolean isOpen = false</p> <p>Before test:</p>  <p>After test:</p>  |
| <b>Pass / Fail</b>     | Pass                                                                                                                                                                                                                                                                                                                                  |

#### Test case 5:

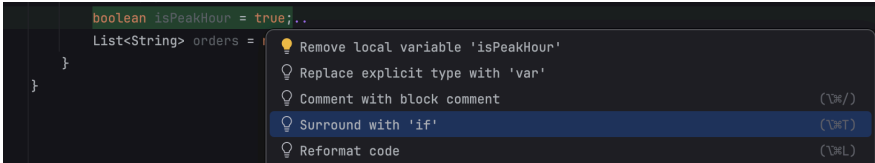
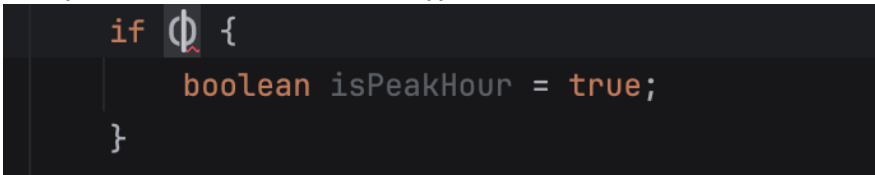
|                    |                                                                                  |
|--------------------|----------------------------------------------------------------------------------|
| <b>Name</b>        | Comment with block comment                                                       |
| <b>Description</b> | Verify that a statement can be wrapped in a block comment via Command Completion |
| <b>Test data</b>   | isOpen = true inside openShop()                                                  |

|                        |                                                                                                                                                                                                                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Steps</b>           | <ol style="list-style-type: none"> <li>1. Place cursor at end of <code>isOpen = true</code></li> <li>2. Type <code>..</code></li> <li>3. Select "Comment with block comment"</li> </ol>  |
| <b>Expected result</b> | Statement is wrapped: <code>/* isOpen = true; */</code>                                                                                                                                                                                                                    |
| <b>Actual result</b>   | <p>Statement wrapped as <code>/*isOpen = true;*/</code> — semicolon included inside comment, result is syntactically valid</p>                           |
| <b>Pass / Fail</b>     | Pass                                                                                                                                                                                                                                                                       |

#### Test case 6:

|                        |                                                                                                                                                                                                                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>            | Reformat code                                                                                                                                                                                                                                                                             |
| <b>Description</b>     | Verify that Reformat code action applies correct formatting via Command Completion                                                                                                                                                                                                        |
| <b>Test data</b>       | Manually add extra spaces: <code>isOpen=true</code> (remove spaces around =)                                                                                                                                                                                                              |
| <b>Steps</b>           | <ol style="list-style-type: none"> <li>1. Break formatting of a line manually</li> <li>2. Place cursor at end of line</li> <li>3. Type <code>..</code></li> <li>4. Select "Reformat code"</li> </ol>  |
| <b>Expected result</b> | Code is reformatted according to IDE code style settings                                                                                                                                                                                                                                  |
| <b>Actual result</b>   | <p>Code reformatted correctly: <code>isOpen=true</code> became <code>isOpen = true</code> with proper spacing according to IDE code style settings</p>                |
| <b>Pass / Fail</b>     | Pass                                                                                                                                                                                                                                                                                      |

### Test case 7:

|                        |                                                                                                                                                                                                                                                                                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>            | Postfix template - .if on boolean                                                                                                                                                                                                                                                                                                          |
| <b>Description</b>     | Verify that .if postfix template wraps boolean variable in an if-statement                                                                                                                                                                                                                                                                 |
| <b>Test data</b>       | isPeakHour boolean variable in main()                                                                                                                                                                                                                                                                                                      |
| <b>Steps</b>           | <ol style="list-style-type: none"> <li>1. Place cursor after isPeakHour</li> <li>2. Type ..</li> <li>3. Select .if</li> </ol>                                                                                                                            |
| <b>Expected result</b> | Code becomes if (isPeakHour) { }                                                                                                                                                                                                                                                                                                           |
| <b>Actual result</b>   | <p>Postfix .if applied incorrectly: entire variable declaration boolean isPeakHour = true; was placed inside the if block body instead of being used as the condition. Result: if () { boolean isPeakHour = true; } instead of if (isPeakHour) { }</p>  |
| <b>Pass / Fail</b>     | Fail                                                                                                                                                                                                                                                                                                                                       |

### Test case 8:

|                        |                                                                                                                                                                                                                                                                                          |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>            | Postfix template — .for on array                                                                                                                                                                                                                                                         |
| <b>Description</b>     | Verify that .for postfix template generates foreach loop for an array                                                                                                                                                                                                                    |
| <b>Test data</b>       | popularItems int array in main()                                                                                                                                                                                                                                                         |
| <b>Steps</b>           | <ol style="list-style-type: none"> <li>1. Place cursor after popularItems</li> <li>2. Type ..</li> <li>3. Select .for</li> </ol>                                                                                                                                                         |
| <b>Expected result</b> | Foreach loop generated: for (int item : popularItems) { }                                                                                                                                                                                                                                |
| <b>Actual result</b>   | <p>Postfix .for applied incorrectly: entire array declaration int[] popularItems = {3, 1, 4, 1, 5} was placed inside the for block body instead of being used as the iterable. Result: for () { int[] popularItems = {3, 1, 4, 1, 5}; } instead of for (int item : popularItems) { }</p> |

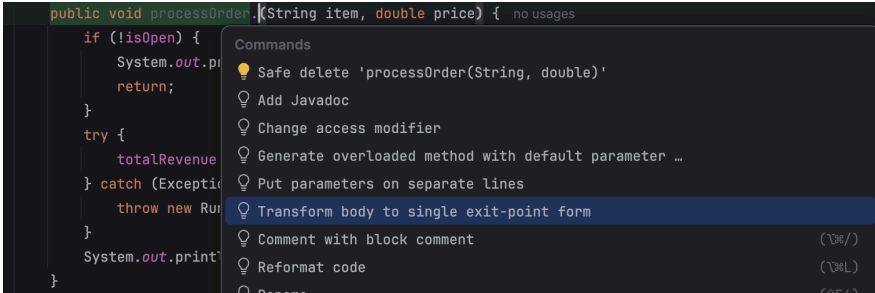
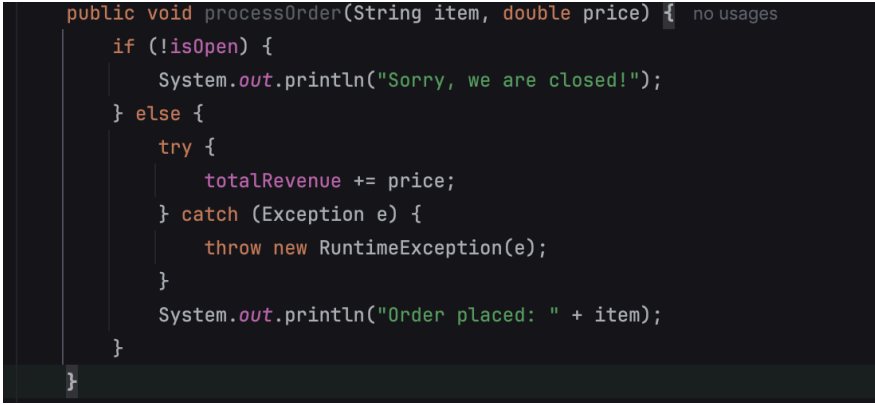
|                    |                                                                                  |
|--------------------|----------------------------------------------------------------------------------|
|                    | <pre> for <math>\phi</math> {     int[] popularItems = {3, 1, 4, 1, 5}; } </pre> |
| <b>Pass / Fail</b> | Fail                                                                             |

#### Test case 9:

|                        |                                                                                                                                                                                                                                  |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>            | AI action — Explain code                                                                                                                                                                                                         |
| <b>Description</b>     | Verify that "Explain code" AI action is available and returns explanation                                                                                                                                                        |
| <b>Test data</b>       | CoffeShop.java                                                                                                                                                                                                                   |
| <b>Steps</b>           | <ol style="list-style-type: none"> <li>1. Open CoffeeShop.java</li> <li>2. Place cursor inside any method</li> <li>3. Type .explain as shown on official documentation</li> <li>4. Select 'Explain code' for dropdown</li> </ol> |
| <b>Expected result</b> | "Explain Code" action appears in dropdown and opens AI explanation panel                                                                                                                                                         |
| <b>Actual result</b>   | Typing .ex does not show "Explain Code" in Command Completion dropdown — only a syntax error is displayed in the editor. Action remains unavailable despite being documented as accessible via .explain shorthand                |
| <b>Pass / Fail</b>     | Fail                                                                                                                                                                                                                             |

#### Test case 10:

|                    |                                                                                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>        | Transform body to single exit-point form                                                                                                                                        |
| <b>Description</b> | Verify that a method with multiple return/exit points is refactored to have a single exit point via Command Completion                                                          |
| <b>Test data</b>   | processOrder(String item, double price) — has early return inside if (!isOpen) block                                                                                            |
| <b>Steps</b>       | <ol style="list-style-type: none"> <li>1. Place cursor on method name processOrder</li> <li>2. Type ..</li> <li>3. Select "Transform body to single exit-point form"</li> </ol> |

|                        |                                                                                                                                                                                                                                                                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        |                                                                                                                                                                                                                                                                 |
| <b>Expected result</b> | Method refactored so that all execution paths lead to a single return statement at the end of the method                                                                                                                                                                                                                                          |
| <b>Actual result</b>   | <p>Method was refactored: early return removed, logic moved into else block. However, the method still has no explicit single return statement — the transformation restructured control flow using if/else instead of achieving a true single exit point</p>  |
| <b>Pass / Fail</b>     | Fail                                                                                                                                                                                                                                                                                                                                              |

### Test case 11:

|                        |                                                                                                                                                                                                                                                                           |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>            | Context sensitivity of Command Completion                                                                                                                                                                                                                                 |
| <b>Description</b>     | Verify that the list of suggested commands changes depending on cursor position within the same method signature                                                                                                                                                          |
| <b>Test data</b>       | processOrder(String item, double price) method                                                                                                                                                                                                                            |
| <b>Steps</b>           | <ol style="list-style-type: none"> <li>1. Place cursor on the method name processOrder → type .. → note the list</li> <li>2. Place cursor after ) before { → type .. → note the list</li> <li>3. Place cursor inside parameter price → type .. → note the list</li> </ol> |
| <b>Expected result</b> | Each cursor position produces a different, contextually relevant list of commands                                                                                                                                                                                         |
| <b>Actual result</b>   | The list of commands changes depending on cursor position: <ol style="list-style-type: none"> <li>1. After name</li> </ol>                                                                                                                                                |

```

public void processOrder(String item, double price) {
    if (!isOpen) {
        System.out.println("Sorry, we are closed!");
    } else {
        try {
            totalRevenue += price;
        } catch (Exception e) {
            throw new RuntimeException(e);
        }
        System.out.println("Order placed: " + item);
    }
}

// public void processOrder(String item, double price) {
//     if (!isOpen) {
//         System.out.println("Sorry, we are closed!");
//         return;
//     }
//     try {
//         totalRevenue += price;
//     } catch (Exception e) {
//         throw new RuntimeException(e);
//     }
// }

```

Commands

- Safe delete 'processOrder(String, double)'
- Add Javadoc
- Change access modifier
- Generate overloaded method with default parameter ...
- Put parameters on separate lines
- Comment with block comment (⌘/)
- Reformat code (⌘L)
- Rename (⌘F6)
- Change signature (⌘F6)
- Move (F6)
- Quick documentation (F1)
- Show usages (⌘F7)
- Copy reference (⌘⌘C)
- Go to members (⌘F12)
- Inline (⌘N)

Press ^ to choose the selected (or first) suggestion and insert a dot afterwards Next Tip

## 2. After some parameter

```

public void processOrder(String item, double price) {
    if (!isOpen) {
        System.out.println("Sorry, we are closed!");
    } else {
        try {
            totalRevenue += price;
        } catch (Exception e) {
            throw new RuntimeException(e);
        }
        System.out.println("Order placed: " + item);
    }
}

// public void processOrder(String item, double price) {
//     if (!isOpen) {
//         System.out.println("Sorry, we are closed!");
//         return;
//     }
// }

```

Commands

- Safe delete 'processOrder(String, double)'
- Generate overloaded method with default parameter ...
- Put parameters on separate lines
- Comment with block comment (⌘/)
- Change signature (⌘F6)
- Reformat code (⌘L)
- Rename (⌘F6)
- Move (F6)
- Show usages (⌘F7)
- Type info (⌘P)
- Go to members (⌘F12)

Press ^ to choose the selected (or first) suggestion and insert a dot afterwards Next Tip

## 3. On {} scope

```

public void processOrder(String item, double price) {
    if (!isOpen) {
        System.out.println("Sorry, we are closed!");
    } else {
        try {
            totalRevenue += price;
        } catch (Exception e) {
            throw new RuntimeException(e);
        }
        System.out.println("Order placed: " + item);
    }
}

```

Commands

- Safe delete 'processOrder(String, double)'
- Comment with block comment (⌘/)
- Reformat code (⌘L)
- Go to members (⌘F12)

Press ^ to choose the selected (or first) suggestion and insert a dot afterwards Next Tip

## 4. Between () and {}

```

public void processOrder(String item, double price) {
    if (!isOpen) {
        System.out.println("Sorry, we are closed!");
    } else {
        try {
            totalRevenue += price;
        } catch (Exception e) {
            throw new RuntimeException(e);
        }
        System.out.println("Order placed: " + item);
    }
}

// public void processOrder(String item, double price) {
//     if (!isOpen) {
//         System.out.println("Sorry, we are closed!");
//         return;
//     }
// }

```

Commands

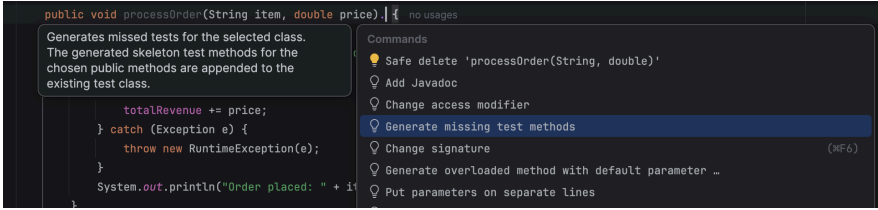
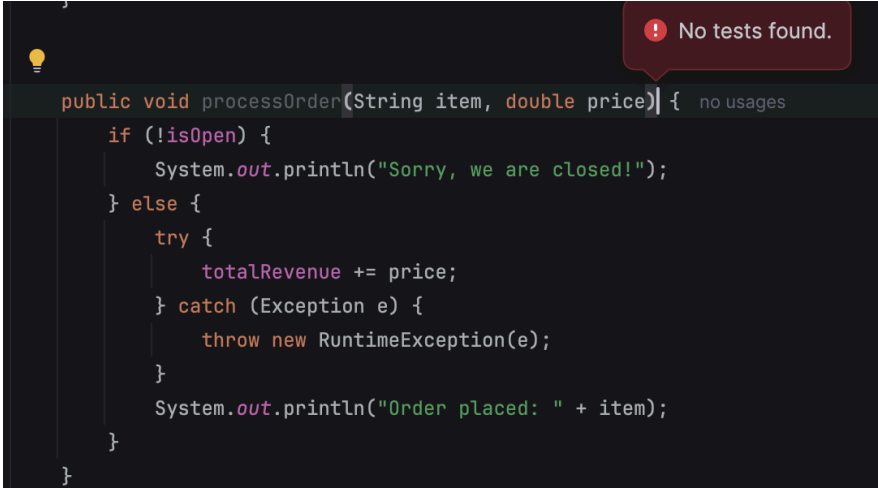
- Safe delete 'processOrder(String, double)'
- Add Javadoc
- Change access modifier
- Generate missing test methods
- Change signature (⌘F6)
- Generate overloaded method with default parameter ...
- Put parameters on separate lines
- Comment with block comment (⌘/)
- Reformat code (⌘L)
- Rename (⌘F6)
- Move (F6)
- Go to members (⌘F12)

Press + to insert, - to replace Next Tip

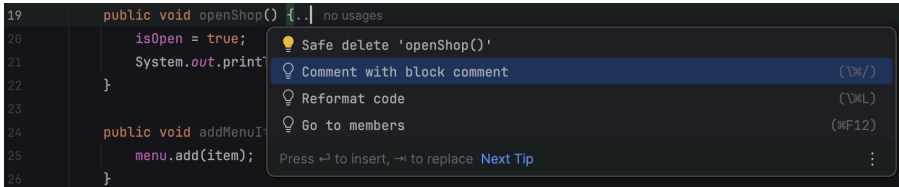
Pass / Fail

Pass

## Test case 12:

|                        |                                                                                                                                                                                                                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>            | Generate missing test methods                                                                                                                                                                                                                                                             |
| <b>Description</b>     | Verify that "Generate missing test methods" action is available and correctly creates a test class with test method stubs                                                                                                                                                                 |
| <b>Test data</b>       | processOrder(String item, double price) method — cursor placed after ) before {                                                                                                                                                                                                           |
| <b>Steps</b>           | <ol style="list-style-type: none"> <li>1. Place cursor after ) in public void processOrder(String item, double price)</li> <li>2. Type ..</li> <li>3. Select "Generate missing test method"</li> </ol>  |
| <b>Expected result</b> | A test class is created with a stub method testProcessOrder() inside                                                                                                                                                                                                                      |
| <b>Actual result</b>   | <p>Error notification "No tests found." displayed — no test class or test method stub was created</p>                                                                                                 |
| <b>Pass / Fail</b>     | Fail                                                                                                                                                                                                                                                                                      |

## Bug 1:

|                            |                                                                                                                                                                                                                                                                                                                                |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Title</b>               | "Comment with block comment" applied to method declaration produces malformed comment syntax                                                                                                                                                                                                                                   |
| <b>Severity</b>            | Major                                                                                                                                                                                                                                                                                                                          |
| <b>Priority</b>            | High                                                                                                                                                                                                                                                                                                                           |
| <b>Connect</b>             | Test case 5                                                                                                                                                                                                                                                                                                                    |
| <b>Environment</b>         | IntelliJ IDEA 2026.1.1, macOS, Java 21                                                                                                                                                                                                                                                                                         |
| <b>Steps to reproduce:</b> | <ol style="list-style-type: none"> <li>1. Open CoffeeShop.java</li> <li>2. Place cursor at the end of method declaration line: public void openShop() {</li> <li>3. Type ..</li> <li>4. Select "Comment with block comment"</li> </ol>       |
| <b>Expected result</b>     | Either the entire method body is wrapped in /* */, OR only the declaration line is commented — behaviour is clear and syntactically valid                                                                                                                                                                                      |
| <b>Actual result</b>       | <p>/*{*/ is inserted into the method signature — comment is placed inside the { brace, producing invalid syntax: public void openShop() /*{*/</p> <p>The IDE cannot determine whether to comment the whole block or just the line, and inserts the comment token in a syntactically incorrect position — breaking the code</p> |
| <b>Photo documentation</b> | Actual state:                                                                                                                                                                                                                                                                                                                  |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> public void openShop() /*{*/ no usages     isOpen = true;     System.out.println(shopName + " is now open!"); }  public void addMenuItem(String item) { no usages     menu.add(item); }  public void processOrder(String item, double price) { no usages     if (!isOpen) {         System.out.println("Sorry, we are closed!");         return;     }     try {         totalRevenue += price;     } catch (Exception e) {         throw new RuntimeException(e);     }     System.out.println("Order placed: " + item); }  public double calculateTax(double taxRate) { no usages     return totalRevenue * taxRate; }  public static void main(String[] args) {     CoffeeShop shop = new CoffeeShop(shopName: "Brew &amp; Code", maxCapacity: 50);     String greeting = "Welcome to our coffee shop";     int[] popularItems = {3, 1, 4, 1, 5};     boolean isPeakHour = true;     List&lt;String&gt; orders = new ArrayList&lt;&gt;(); } } </pre> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Bug 2:

|                        |                                                                                                                                                                                 |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Title</b>           | "Transform body to single exit-point form" does not produce a single exit point for void method with early return                                                               |
| <b>Connect</b>         | Test - case 10                                                                                                                                                                  |
| <b>Severity</b>        | Major                                                                                                                                                                           |
| <b>Priority</b>        | Medium                                                                                                                                                                          |
| <b>Description</b>     | Verify that Transform body to single exit-point form correctly refactors method control flow                                                                                    |
| <b>Steps</b>           | <ol style="list-style-type: none"> <li>1. Place cursor on processOrder method name</li> <li>2. Type ..</li> <li>3. Select "Transform body to single exit-point form"</li> </ol> |
| <b>Expected result</b> | All execution paths merged into one — single return at the end of the method                                                                                                    |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actual result</b> | <p>Early return removed but code restructured into if/else — there is still no explicit single exit point, just different control flow structure</p> <pre> /**  * @param item  * @param price  */ public void processOrder(String item, double price) { no usages     if (!isOpen) {         System.out.println("Sorry, we are closed!");         return;     }     try {         totalRevenue += price;     } catch (Exception e) {         throw new RuntimeException(e);     }     System.out.println("Order placed: " + item); }  public double calculateTax(double taxRate) { no usages </pre> |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Bug 3;

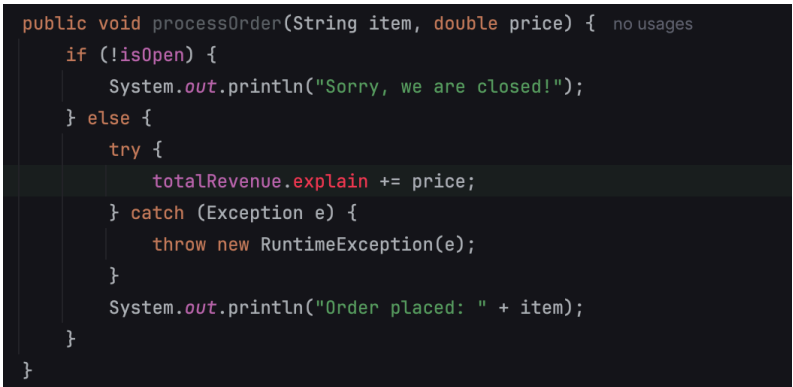
|                        |                                                                                                                                                                                       |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Title</b>           | Postfix template .if wraps entire variable declaration into if-body instead of using variable as condition                                                                            |
| <b>Connect</b>         | Test - case 7                                                                                                                                                                         |
| <b>Severity</b>        | Major                                                                                                                                                                                 |
| <b>Priority</b>        | High                                                                                                                                                                                  |
| <b>Steps</b>           | <ol style="list-style-type: none"> <li>1. Declare boolean isPeakHour = true in main()</li> <li>2. Place cursor after isPeakHour</li> <li>3. Type ..</li> <li>4. Select .if</li> </ol> |
| <b>Expected result</b> | if (isPeakHour) { } — variable used as condition                                                                                                                                      |
| <b>Actual result</b>   | if () { boolean isPeakHour = true; } — entire declaration moved into if-body, condition left empty                                                                                    |

### Bug 4;

|                 |                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------|
| <b>Title</b>    | Postfix template .for wraps entire array declaration into for-body instead of generating foreach loop |
| <b>Connect</b>  | Test - case 8                                                                                         |
| <b>Severity</b> | Major                                                                                                 |

|                        |                                                                                                                                                                                                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Priority</b>        | High                                                                                                                                                                                                                                                                 |
| <b>Steps</b>           | <ol style="list-style-type: none"> <li>1. Declare <code>int[] popularItems = {3, 1, 4, 1, 5}</code> in <code>main()</code></li> <li>2. Place cursor after <code>popularItems</code></li> <li>3. Type <code>..</code></li> <li>4. Select <code>.for</code></li> </ol> |
| <b>Expected result</b> | <code>for (int item : popularItems) { }</code> — foreach loop generated with array as iterable                                                                                                                                                                       |
| <b>Actual result</b>   | <code>for () { int[] popularItems = {3, 1, 4, 1, 5}; }</code> — entire declaration moved into for-body, loop header left empty                                                                                                                                       |

## Bug 5:

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Title</b>           | "Explain code" AI action is not available via Command Completion despite being documented                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Connect</b>         | Test - case 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Description</b>     | According to the Task #1 description, new AI-powered actions (Explain code, Generate documentation) should be discoverable and invokable via <code>..</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Severity</b>        | Major                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Priority</b>        | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Steps</b>           | <ol style="list-style-type: none"> <li>1. Open <code>CoffeeShop.java</code></li> <li>2. Place cursor inside any method</li> <li>3. Type <code>.explain</code> as shown on official documentation</li> <li>4. Select 'Explain code' for dropdown</li> </ol>  <pre> public void processOrder(String item, double price) { no usages     if (!isOpen) {         System.out.println("Sorry, we are closed!");     } else {         try {             totalRevenue.explain += price;         } catch (Exception e) {             throw new RuntimeException(e);         }         System.out.println("Order placed: " + item);     } } </pre> |
| <b>Expected result</b> | "Explain Code" action appears in dropdown and opens AI explanation panel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Actual result</b>   | Typing <code>.explain</code> does not show "Explain Code" in Command Completion dropdown — only a syntax error is displayed in the editor. Action remains unavailable despite being documented as accessible via <code>.ex</code> shorthand                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|  |                                                                                                                                                                                                                                                                                                                                                                            |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> public void openShop() {<u>explain</u> no usages     isOpen = true;     System.out.println(shopName + " is now open!"); }  public class CoffeeShop {     private String <u>shopName</u>.<u>explain</u>; 2 usages     private int <u>maxCapacity</u>; 1 usage     private boolean <u>isOpen</u>; 3 usages     private List&lt;String&gt; <u>menu</u>; 2 usages </pre> |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Bug 6:

|                            |                                                                                                                                                                                                                      |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Title</b>               | "Generate missing test methods" shows "No tests found" error instead of creating test stubs                                                                                                                          |
| <b>Connect</b>             | Test case 12                                                                                                                                                                                                         |
| <b>Severity</b>            | Major                                                                                                                                                                                                                |
| <b>Priority</b>            | High                                                                                                                                                                                                                 |
| <b>Environment</b>         | IntelliJ IDEA 2026.1.1, macOS, Java 21                                                                                                                                                                               |
| <b>Steps to reproduce:</b> | <ol style="list-style-type: none"> <li>1. Place cursor after ) and before { in public void processOrder(String item, double price)</li> <li>2. Type ..</li> <li>3. Select "Generate missing test methods"</li> </ol> |
| <b>Expected result</b>     | Test class created with stub method testProcessOrder() inside                                                                                                                                                        |
| <b>Actual result</b>       | Error popup "No tests found." appears — no test class or stub is generated                                                                                                                                           |
| <b>Possible cause</b>      | No test framework (e.g. JUnit) configured in the project. Expected behavior: IDE should either prompt to add a test framework or display a clearer error message explaining the requirement.                         |

## Additional bugs

### Additional bug:

|                 |                                                                                     |
|-----------------|-------------------------------------------------------------------------------------|
| <b>Title</b>    | "Add Javadoc" generates empty Javadoc block without method description              |
| <b>Severity</b> | Major <ul style="list-style-type: none"> <li>- Placeholder functionality</li> </ul> |

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Priority        | Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Connect         | Connect: Discovered during exploratory testing (related to TC11 — cursor on method name)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Description     | Verify that JavaDoc is generated for a method via Command Completion with relevant description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Test data       | processOrder(String item, double price) method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Steps           | <ol style="list-style-type: none"> <li>1. Place cursor on method name processOrder</li> <li>2. Type . right after the name</li> <li>3. Select "add javadoc"</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Expected result | JavaDoc generated with @param tags <b>and</b> a meaningful method description above them                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Actual result   | <p>JavaDoc block generated with @param item and @param price, but <b>without any description</b> — the comment body is empty, no method summary provided</p> <pre> /**  * @param item  * @param price  */ public void processOrder(String item, double price) { no usages     if (!isOpen) {         System.out.println("Sorry, we are closed!");         return;     }     try {         totalRevenue += price;     } catch (Exception e) {         throw new RuntimeException(e);     }     System.out.println("Order placed: " + item); }  public double calculateTax(double taxRate) { no usages </pre> |
| Pass / Fail     | FAIL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## Testing summary

|                  |         |
|------------------|---------|
| Total test cases | 12      |
| Passed           | 7       |
| Failed           | 5       |
| Total bug found  | 7       |
| Testing time     | 3 hours |

## Further Testing Areas

### Areas for deeper testing:

- **Kotlin support** — all tested cases repeated for Kotlin files
- **.properties files** — Command Completion behavior in configuration files (new in 2026.1.1)
- **Undo/Redo** — verify all commands are fully reversible
- **Invocation inside comments and String literals** — edge case behavior
- **Performance** — dropdown response time on large files (1000+ lines)
- **All remaining postfix templates** — .null, .nonnull, .not, .return, .throw, .cast
- **AI actions** — "Generate documentation" and "Make code changes" via prompt dialog
- **Keyboard navigation** — using arrow keys and Enter instead of mouse click
- **Command search/filter** — typing to filter commands inside the dropdown
- **Comment behavior in different code locations** — testing "Comment with block comment" action when cursor is placed inside a String literal, inside an existing comment, on a class declaration, on an import statement, and on an annotation to verify correct behavior in each context